

OMRON E2E NEXT SERIES INDUCTIVE PROXIMITY SENSOR
TECHNICAL SPECIFICATION DATASHEET - REF: E2E-X8MB1D18

1. SENSING CHARACTERISTICS

Model Identifier: E2E-X8MB1D18 2M

Sensor Family: E2E NEXT High-Performance Inductive Sensors

Housing Size: M18 Cylindrical Barrel (Thread length: 49 mm)

Installation / Shielding: Shielded / Flush Mountable

Rated Sensing Distance: 8.0 mm (+/- 10%)

Stable Sensing Distance: 0 to 6.4 mm

Standard Target Object: Mild Steel 24 x 24 x 1 mm

Differential Travel (Hysteresis): 10% max of sensing distance

Response Switching Frequency: 1.5 kHz (1500 operations/sec)

2. ELECTRICAL RATINGS & OUTPUT CIRCUITS

Power Supply Voltage: 10 to 30 VDC (including 10% ripple p-p)

Operating Voltage Range: 10 to 32 VDC

Current Consumption: 10 mA max @ 24 VDC (no-load)

Control Output Type: DC 3-Wire PNP Open Collector

Output Operation Mode: Normally Open (NO)

Maximum Load Current: 200 mA max @ 30 VDC

Residual Voltage: 2.0 V max @ load current 200 mA

Circuit Protection: Reverse polarity protection, Surge suppressor, Output short-circuit protection

Indicator LEDs: Operation indicator (Orange LED), Stable operation indicator (Green LED)

3. ENVIRONMENTAL & MECHANICAL TOLERANCES

Enclosure Protection Rating: IP67, IP69K (Oil-resistant according to DIN 40050-9)

Ambient Operating Temperature: -40 C to +85 C (No icing or condensation)

Ambient Operating Humidity: 35% to 95% RH

Insulation Resistance: 50 MOhm min @ 500 VDC between energized parts and case

Dielectric Strength: 1000 VAC, 50/60 Hz for 1 minute

Vibration Resistance: 10 to 55 Hz, 1.5 mm double amplitude for 2 hours each in X, Y, Z directions

Shock Resistance: 1000 m/s² (approx 100G) 10 times each in X, Y, Z directions

Housing Material: Brass Nickel-Plated

Sensing Face Material: PBT (Polybutylene Terephthalate)

Connection Type: Pre-wired Oil-Resistant PVC Cable (2 meters length, 3-conductor)